

Higgs Induced Signal in Cosmology

Abstract

Contents

1	Introduction	3
2	Gauge Boson Signal	3
2.1	Preliminaries	3
2.2	Bubble Diagram	6
2.3	Triangle Diagram	8
2.4	Box Diagram	11
3	Top Quark Signal	15
4	Conclusions	15
A	Loop Integral Formulae	15
B	Bubble Coefficients	15
C	Triangle Coefficients	16
C.1	Polarization sum	16
C.2	Time-integral coefficients	17
D	Box Coefficients	18
D.1	Polarization sums	18
D.2	Angular projections	20

1 Introduction

2 Gauge Boson Signal

2.1 Preliminaries

We will compute the connected four-point function $\langle \delta h_{\mathbf{k}_1} \delta h_{\mathbf{k}_2} \delta h_{\mathbf{k}_3} \delta h_{\mathbf{k}_4} \rangle'$ generated by a massive gauge-boson loop. Our interest is the leading nonlocal signal in the collapsed s channel,

$$\mathbf{k}_s \equiv \mathbf{k}_1 + \mathbf{k}_2 = -(\mathbf{k}_3 + \mathbf{k}_4), \quad k_s \ll k_i, \quad k_{12} \equiv k_1 + k_2, \quad k_{34} \equiv k_3 + k_4, \quad (1)$$

where $k_i \equiv |\mathbf{k}_i|$. Thus $k_1 \simeq k_2$ and $k_3 \simeq k_4$ at leading order. We choose $\mathbf{k}_s$ along the z axis and set the azimuth of $\mathbf{k}_1$ to zero. Here θ_1 and θ_3 are the polar angles of $\mathbf{k}_1$ and $\mathbf{k}_3$, respectively, and ϕ_3 is their relative azimuth.

The Higgs–gauge couplings follow from the Higgs kinetic term [1]:

$$\frac{\sqrt{-g}}{2} |D_\mu \mathbf{H}|^2 \supset \frac{a^2 g^2}{4} (h_0 + \delta h)^2 \left(W_\mu^+ W^{-\mu} + \frac{Z_\mu Z^\mu}{2c_W^2} \right), \quad (2)$$

where g is the weak gauge coupling, $c_W = \cos \theta_W$, and the Lorentz contractions on the right use the flat conformal metric. The terms linear and quadratic in δh generate the cubic and quartic vertices, respectively. They give the bubble, two triangle placements, and box shown in Fig. 1.

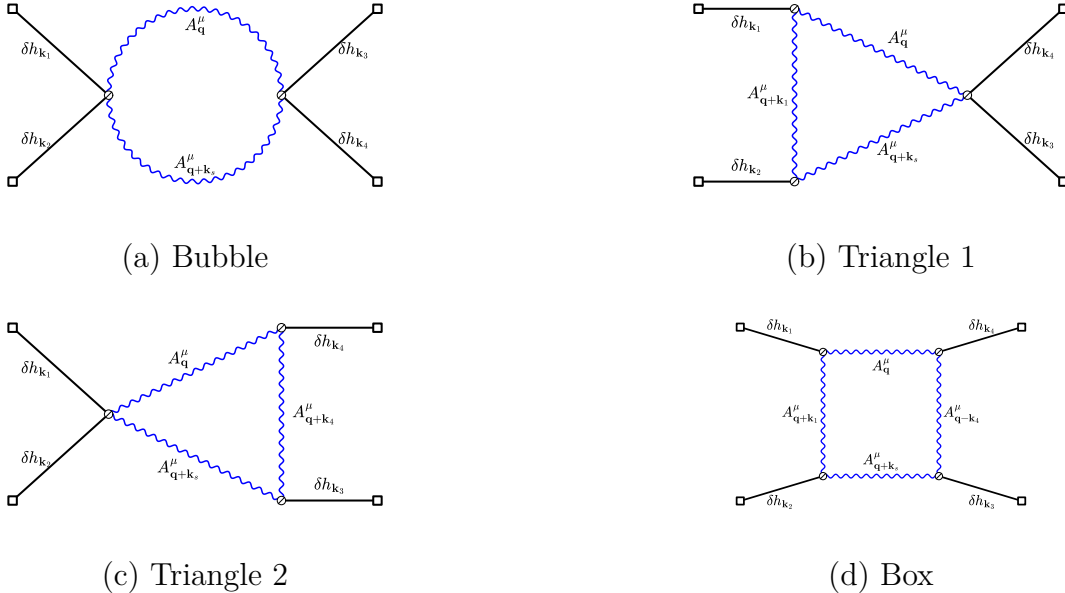


Figure 1: The four gauge-boson loop diagrams considered here. Black solid lines denote external Higgs modes and blue wavy lines denote gauge-boson propagators.

In the light-Higgs regime, the dominant infrared Higgs-loop contribution to the vector masses is [1]

$$m_W^2 \simeq \frac{3g^2 H^4}{8\pi^2 m_H^2}, \quad m_Z^2 \simeq \frac{m_W^2}{c_W^2}. \quad (3)$$

Here m_H is the effective Higgs mass. We use a constant vector mass m_V and the parameter $\tilde{\nu} \equiv \sqrt{m_V^2/H^2 - 1/4} > 0$ in the following. The W and Z contributions are obtained with their respective masses and couplings in (2).

We use $\lambda, \sigma = L, +, -, T$ for vector polarization-basis labels, and $\eta = \pm 1$ for the two nonlocal branches. Neglecting the Higgs mass in the external mode functions, we write its bulk-to-boundary propagator as

$$G_a(k; \tau) = \frac{H^2}{2k^3} (1 - iak\tau) e^{iak\tau}. \quad (4)$$

The spacetime vector propagator decomposes as

$$\begin{aligned} D_{ab,\mu\nu}(\mathbf{k}; \tau_1, \tau_2) = & \sum_{\lambda=T,L,\pm} e_{\mathbf{k},\mu}^{(\lambda)} e_{\mathbf{k},\nu}^{(\lambda)*} D_{ab}^{(\lambda)}(k; \tau_1, \tau_2) \\ & + e_{\mathbf{k},\mu}^{(T)} e_{\mathbf{k},\nu}^{(L)*} D_{ab}^{(TL)}(k; \tau_1, \tau_2) + e_{\mathbf{k},\mu}^{(L)} e_{\mathbf{k},\nu}^{(T)*} D_{ab}^{(LT)}(k; \tau_1, \tau_2), \end{aligned} \quad (5)$$

where $\mu, \nu = 0, 1, 2, 3$. In the comoving frame, the covariant polarization basis is [2]

$$e_{\mathbf{k},\mu}^{(T)} = (1, \mathbf{0}), \quad e_{\mathbf{k},\mu}^{(L)} = (0, \hat{\mathbf{k}}), \quad e_{\mathbf{k},\mu}^{(\pm)} = \frac{1}{\sqrt{2}} (0, \mathbf{e}_\theta \pm i\mathbf{e}_\phi),$$

with $(\hat{\mathbf{k}}, \mathbf{e}_\theta, \mathbf{e}_\phi)$ the orthonormal spherical basis. The label T denotes the temporal component of the longitudinal mode.

At zero chemical potential, all components can be expressed through an auxiliary scalar propagator with the same Hankel index [2, 3]:

$$D_{>}^{(s)}(k; \tau_1, \tau_2) = \frac{\pi H^2}{4} e^{-\pi\tilde{\nu}} (\tau_1 \tau_2)^{3/2} H_{i\tilde{\nu}}^{(1)}(-k\tau_1) H_{-i\tilde{\nu}}^{(2)}(-k\tau_2), \quad (6)$$

$$D_{>}^{(\pm)} = \frac{D_{>}^{(s)}}{H^2 \tau_1 \tau_2}, \quad D_{>}^{(L)} = \frac{(\partial_{\tau_1} - 2/\tau_1)(\partial_{\tau_2} - 2/\tau_2) D_{>}^{(s)}}{H^2(\tilde{\nu}^2 + 1/4)}, \quad (7)$$

$$D_{>}^{(T)} = \frac{k^2 D_{>}^{(s)}}{H^2(\tilde{\nu}^2 + 1/4)}, \quad (8)$$

$$D_{>}^{(TL)} = \frac{ik(\partial_{\tau_2} - 2/\tau_2) D_{>}^{(s)}}{H^2(\tilde{\nu}^2 + 1/4)}, \quad D_{>}^{(LT)} = -\frac{ik(\partial_{\tau_1} - 2/\tau_1) D_{>}^{(s)}}{H^2(\tilde{\nu}^2 + 1/4)}. \quad (9)$$

With spacetime indices and momentum arguments suppressed, $D_{-+} = D_{>}$, $D_{+-} = D_{<}$, and the separated-time SK propagators are

$$\begin{aligned} D_{++} &= \theta(\tau_1 - \tau_2) D_{>} + \theta(\tau_2 - \tau_1) D_{<}, \\ D_{--} &= \theta(\tau_2 - \tau_1) D_{>} + \theta(\tau_1 - \tau_2) D_{<}. \end{aligned} \quad (10)$$

These relations apply to $D^{(\pm)}, D^{(L)}, D^{(TL)}$ and $D^{(LT)}$. But for $D^{(T)}$, the $++$ and $--$ propagators have a delta-function term because the temporal mode is not a physical degree of freedom:

$$\begin{aligned} D_{++}^{(T)} &= \theta(\tau_1 - \tau_2) D_{>}^{(T)} + \theta(\tau_2 - \tau_1) D_{<}^{(T)} + \frac{i\delta(\tau_1 - \tau_2)}{H^2(\tilde{\nu}^2 + 1/4)a(\tau_1)^2}, \\ D_{--}^{(T)} &= \theta(\tau_2 - \tau_1) D_{>}^{(T)} + \theta(\tau_1 - \tau_2) D_{<}^{(T)} - \frac{i\delta(\tau_1 - \tau_2)}{H^2(\tilde{\nu}^2 + 1/4)a(\tau_1)^2}. \end{aligned} \quad (11)$$

Then we will see that the propagators $D^{(T)}$, $D^{(TL)}$ and $D^{(LT)}$ will not contribute to the leading nonlocal signal.

The leading nonlocal parts of the spatial propagators are independent of the SK indices:

$$\left[D_{\text{ab}}^{(\pm)}(k; \tau_1, \tau_2) \right]_{\text{NonL}} = \frac{(\tau_1 \tau_2)^{1/2}}{4\pi} \sum_{\eta=\pm 1} \Gamma^2(-i\eta\tilde{\nu}) \left(\frac{k^2 \tau_1 \tau_2}{4} \right)^{i\eta\tilde{\nu}}, \quad (12)$$

$$\left[D_{\text{ab}}^{(L)}(k; \tau_1, \tau_2) \right]_{\text{NonL}} = \frac{(\tau_1 \tau_2)^{1/2}}{4\pi} \sum_{\eta=\pm 1} \frac{(-1/2 + i\eta\tilde{\nu})^2}{\tilde{\nu}^2 + 1/4} \Gamma^2(-i\eta\tilde{\nu}) \left(\frac{k^2 \tau_1 \tau_2}{4} \right)^{i\eta\tilde{\nu}}. \quad (13)$$

The temporal nonlocal part is

$$\left[D_{\text{ab}}^{(T)}(k; \tau_1, \tau_2) \right]_{\text{NonL}} = \frac{k^2 (\tau_1 \tau_2)^{3/2}}{4\pi(\tilde{\nu}^2 + 1/4)} \sum_{\eta=\pm 1} \Gamma^2(-i\eta\tilde{\nu}) \left(\frac{k^2 \tau_1 \tau_2}{4} \right)^{i\eta\tilde{\nu}}. \quad (14)$$

The one-loop factorization theorem and cutting rule [4] isolate the leading nonlocal signal in the collapsed channel. We first identify the two internal lines whose removal separates the external pairs (1, 2) and (3, 4). Their momenta can be chosen as $\mathbf{q}$ and $\mathbf{K} \equiv \mathbf{q} + \mathbf{k}_s$. The relevant loop region has $q, K \sim k_s \ll k_i$. We replace these two propagators by the leading nonlocal terms of their late-time expansions, keeping their polarization tensors, and expand the remaining hard propagators to leading order in the soft momenta. The hard propagators retain their full time dependence and SK ordering.

The cut propagators are independent of the SK indices, and their time dependence separates between the two endpoints. The time integrals thus factorize into two tree subgraphs with additional time powers supplied by the cut lines. Multiplying these subgraphs by the soft momentum integral and summing over polarizations gives the nonlocal signal. The branches $\eta = \pm 1$ combine into the oscillatory signal retained below. Higher terms in the soft expansion give subleading corrections in the collapsed limit.

The propagators $D^{(T)}$, $D^{(TL)}$, and $D^{(LT)}$ do not contribute to the leading nonlocal signal. On a soft line, each temporal mode function $B^{(T)}$ carries an additional factor of $k\tau$ relative to the spatial modes $B^{(L)}$ and $B^{(\pm)}$. Their nonanalytic contributions are therefore suppressed in the collapsed limit. The contact term in $D^{(T)}$ is analytic in the soft momentum and does not contribute to the nonlocal signal. On a hard line, a temporal endpoint has a vanishing polarization contraction with the spatial components of the attached soft line. A nonzero contraction would require a temporal soft component, which is already suppressed. We therefore omit propagators with temporal components when computing the leading signal below.

All required soft momentum integrals reduce to five dimensionless functions f_r , $r = 0, 1, 2, 3, 4$. With $z \equiv \hat{\mathbf{q}} \cdot \hat{\mathbf{k}}_s$, we define

$$\int \frac{d^3 \mathbf{q}}{(2\pi)^3} q^{-2\alpha} K^{-2\beta} z^r = \frac{k_s^{3-2\alpha-2\beta}}{(4\pi)^{3/2}} f_r(\alpha, \beta), \quad r = 0, 1, 2, 3, 4. \quad (15)$$

These expressions are understood by analytic continuation from the convergent domain. The first

three functions are

$$f_0(\alpha, \beta) = \frac{\Gamma(\alpha + \beta - 3/2)\Gamma(3/2 - \alpha)\Gamma(3/2 - \beta)}{\Gamma(3 - \alpha - \beta)\Gamma(\alpha)\Gamma(\beta)}, \quad (16)$$

$$f_1(\alpha, \beta) = \frac{1}{2} \left[f_0(\alpha + \frac{1}{2}, \beta - 1) - f_0(\alpha - \frac{1}{2}, \beta) - f_0(\alpha + \frac{1}{2}, \beta) \right], \quad (17)$$

$$f_2(\alpha, \beta) = \frac{1}{4} [f_0(\alpha + 1, \beta - 2) + f_0(\alpha - 1, \beta) + f_0(\alpha + 1, \beta) - 2f_0(\alpha, \beta - 1) - 2f_0(\alpha + 1, \beta - 1) + 2f_0(\alpha, \beta)]. \quad (18)$$

The functions f_3 and f_4 are likewise finite sums of f_0 ; their explicit expressions are collected in App. A.

For the nonlocal branch η , it is convenient to set $x = -i\eta\tilde{\nu}$ and define

$$\mathcal{J}_r(m, n; x) \equiv f_r \left(x - \frac{m}{2}, x + \frac{n}{2} \right), \quad r = 0, 1, 2, 3, 4. \quad (19)$$

Here m, n are integer shifts of momentum powers, unrelated to m_V . In particular,

$$\int \frac{d^3\mathbf{q}}{(2\pi)^3} \left(\frac{q}{2} \right)^{-2x} \left(\frac{K}{2} \right)^{-2x} q^m K^{-n} z^r = \frac{4^{2x} k_s^{3-4x+m-n}}{(4\pi)^{3/2}} \mathcal{J}_r(m, n; x). \quad (20)$$

2.2 Bubble Diagram

The bubble contains two quartic vertices, each attached to a pair of external Higgs modes. With the four external momenta understood, the s -channel contribution is

$$\begin{aligned} \langle \delta h^4 \rangle'_{\text{Bub}} &= \frac{g^4}{8} \sum_{\mathbf{a}, \mathbf{b}=\pm 1} (-\mathbf{ab}) \int_{-\infty}^0 \frac{d\tau_1}{|H\tau_1|^2} \int_{-\infty}^0 \frac{d\tau_2}{|H\tau_2|^2} G_{\mathbf{a}}(k_1; \tau_1) G_{\mathbf{a}}(k_2; \tau_1) G_{\mathbf{b}}(k_3; \tau_2) G_{\mathbf{b}}(k_4; \tau_2) \\ &\times \int \frac{d^3\mathbf{q}}{(2\pi)^3} D_{\mathbf{ab},ij}(\mathbf{q}; \tau_1, \tau_2) D_{\mathbf{ba},ji}(\mathbf{K}; \tau_2, \tau_1), \quad \mathbf{K} = \mathbf{q} + \mathbf{k}_s. \end{aligned} \quad (21)$$

The symmetry factor $1/2$ is included. Throughout this section, the $W^\pm$ contribution is obtained by multiplying the expressions by 2, and the Z contribution by making the replacement $g \rightarrow g/c_W$, with the corresponding mass used in each case.

Replacing the two soft propagators by (12)–(13) removes their SK-index dependence, as illustrated in Fig. 2. After summing the polarizations, the oscillatory part of the loop integral is

$$\begin{aligned} &\left[\int \frac{d^3\mathbf{q}}{(2\pi)^3} D_{\mathbf{ab},ij}(\mathbf{q}; \tau_1, \tau_2) D_{\mathbf{ba},ji}(\mathbf{K}; \tau_2, \tau_1) \right]_{\text{NonL}} \\ &\simeq \sum_{\eta=\pm 1} (\tau_1 \tau_2)^{1+2i\eta\tilde{\nu}} \frac{\Gamma^4(-i\eta\tilde{\nu})}{(4\pi)^{7/2}} 4^{-2i\eta\tilde{\nu}} k_s^{3+4i\eta\tilde{\nu}} \mathcal{I}_{\text{Bub}}(-i\eta\tilde{\nu}). \end{aligned} \quad (22)$$

The dimensionless polarization sum has the compact form

$$\mathcal{I}_{\text{Bub}}(x) = \frac{3+2x}{1-2x} \mathcal{J}_0(0, 0; x) + \frac{16x^2}{(1-2x)^2} [\mathcal{J}_0(2, 2; x) + 2\mathcal{J}_1(1, 2; x) + \mathcal{J}_2(0, 2; x)], \quad (23)$$

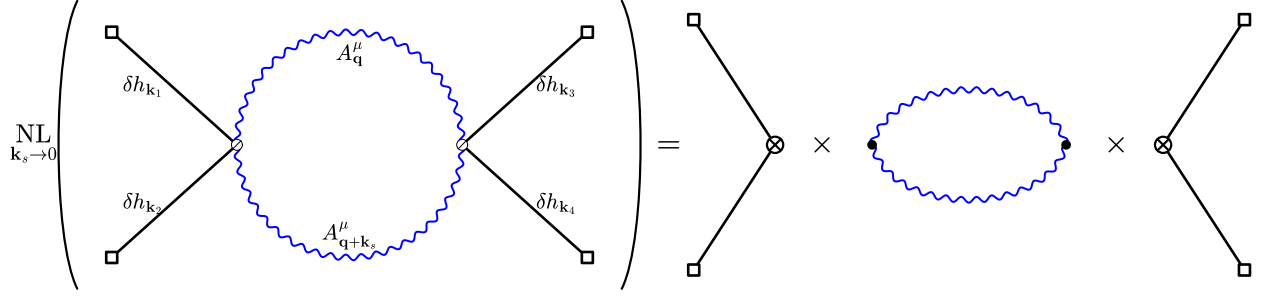


Figure 2: The leading nonlocal part of the bubble in the $k_s \rightarrow 0$ limit. The two soft gauge-boson lines supply the central loop factor, while the left and right subgraphs give the two quartic-vertex time integrals. Crossed circles mark the endpoints of the cut lines in the tree subgraphs.

with $x = -i\eta\tilde{\nu}$ as in (19). The separate polarization contributions are given in App. B.

The remaining time integrals factorize. For the left subgraph, at $k_1 \simeq k_2 \simeq k_{12}/2$, we find

$$\sum_{\mathbf{a}=\pm 1} i\mathbf{a} \int_{-\infty}^0 \frac{d\tau}{|H\tau|^2} G_{\mathbf{a}}(k_1; \tau) G_{\mathbf{a}}(k_2; \tau) (-\tau)^{1+2i\eta\tilde{\nu}} \simeq \frac{H^2 k_{12}^{-2i\eta\tilde{\nu}}}{4k_1^3 k_2^3} \mathcal{P}_{\eta}^{(1)}(\tilde{\nu}), \quad (24)$$

$$\mathcal{P}_{\eta}^{(1)}(\tilde{\nu}) = \frac{2 + i\eta\tilde{\nu}}{2i\eta\tilde{\nu}} i \sinh(\pi\eta\tilde{\nu}) \Gamma(2 + 2i\eta\tilde{\nu}). \quad (25)$$

The right vertex gives the same expression with $(k_1, k_2, k_{12}) \rightarrow (k_3, k_4, k_{34})$. Combining the two vertices with (22), we obtain

$$\left[\langle \delta h^4 \rangle'_{\text{Bub}} \right]_{\text{NonL}} \simeq \frac{g^4 H^4 k_s^3}{128 k_1^3 k_2^3 k_3^3 k_4^3} \left[\left(\frac{k_s^2}{4k_{12}k_{34}} \right)^{2i\tilde{\nu}} C^{\text{Bub}}(\tilde{\nu}) + \text{c.c.} \right], \quad (26)$$

where the complex signal coefficient is

$$C^{\text{Bub}}(\tilde{\nu}) = \frac{\Gamma^4(-i\tilde{\nu})}{(4\pi)^{7/2}} \left[\mathcal{P}_+^{(1)}(\tilde{\nu}) \right]^2 \mathcal{I}_{\text{Bub}}(-i\tilde{\nu}). \quad (27)$$

This definition separates the loop and time integrals from the couplings and diagram symmetry factor. The bracket in (26) equals

$$2|C^{\text{Bub}}(\tilde{\nu})| \cos \left[2\tilde{\nu} \log \left(\frac{k_s^2}{4k_{12}k_{34}} \right) + \arg C^{\text{Bub}}(\tilde{\nu}) \right]. \quad (28)$$

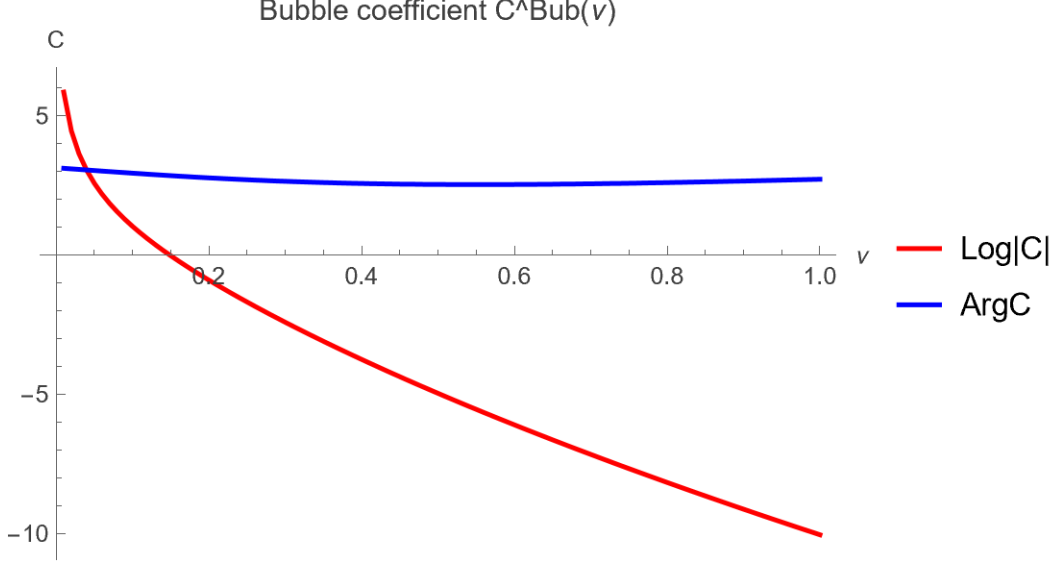


Figure 3: The bubble coefficient defined in (27). The red and blue curves show $\log |C^{Bub}|$ and $\arg C^{Bub}$ in radians, respectively, as functions of $\tilde{\nu}$.

2.3 Triangle Diagram

The triangle contains two cubic vertices and one quartic vertex. We first consider the diagram “Triangle 1”. The triangle has a unit diagram symmetry factor. In the conventions of Sec. 2.2, its integral expression is

$$\begin{aligned} \langle \delta h^4 \rangle'_{\text{Tri1}} &= \frac{g^6 h_0^2}{8} \sum_{a,b,c=\pm 1} (-iabc) \int_{-\infty}^0 \prod_{v=1}^3 \frac{d\tau_v}{|H\tau_v|^2} G_a(k_1; \tau_1) G_b(k_2; \tau_2) G_c(k_3; \tau_3) G_c(k_4; \tau_3) \\ &\quad \times \int \frac{d^3 \mathbf{q}}{(2\pi)^3} D_{ab,ij}(\mathbf{q} + \mathbf{k}_1; \tau_1, \tau_2) D_{bc,j\ell}(\mathbf{K}; \tau_2, \tau_3) D_{ca,\ell i}(\mathbf{q}; \tau_3, \tau_1). \end{aligned} \quad (29)$$

The nonlocal signal comes from $q, K \sim k_s \ll k_1$. We cut the $\mathbf{q}$ and $\mathbf{K}$ lines and evaluate the remaining hard propagator at $\mathbf{q} + \mathbf{k}_1 \simeq \mathbf{k}_1$. The latter retains its full time dependence and SK ordering. The resulting factorization is shown in Fig. 4.

After summing the soft polarizations, the cut loop takes the form

$$\begin{aligned} &\left[\int \frac{d^3 \mathbf{q}}{(2\pi)^3} D_{ab,ij}(\mathbf{q} + \mathbf{k}_1; \tau_1, \tau_2) D_{bc,j\ell}(\mathbf{K}; \tau_2, \tau_3) D_{ca,\ell i}(\mathbf{q}; \tau_3, \tau_1) \right]_{\text{NonL}} \\ &\simeq \sum_{\lambda=L,\pm} \sum_{\eta=\pm 1} D_{ab}^{(\lambda)}(k_1; \tau_1, \tau_2) (\tau_1 \tau_2)^{1/2+i\eta\tilde{\nu}} (-\tau_3)^{1+2i\eta\tilde{\nu}} \frac{\Gamma^4(-i\eta\tilde{\nu})}{(4\pi)^{7/2}} 4^{-2i\eta\tilde{\nu}} k_s^{3+4i\eta\tilde{\nu}} \mathcal{I}_{\text{Tri}}^\lambda(-i\eta\tilde{\nu}, \theta_1). \end{aligned} \quad (30)$$

This equation defines the dimensionless coefficients $\mathcal{I}_{\text{Tri}}^\lambda(x, \theta)$, with $x = -i\eta\tilde{\nu}$. After simplification, we surprisingly find that all three coefficients equal $\mathcal{I}_{\text{Bub}}(x)/3$ and the apparent angular dependence cancels, as shown in (51).

We next evaluate the left and right subgraphs. The quartic vertex gives the time integral $\mathcal{P}_\eta^{(1)}$

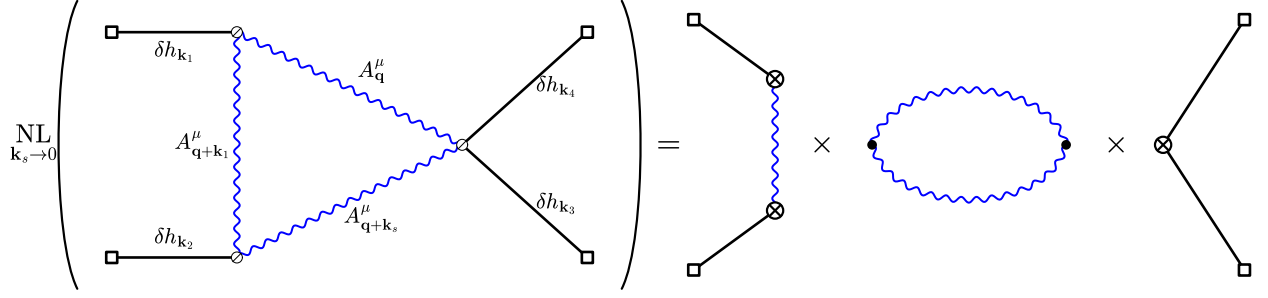


Figure 4: The leading nonlocal part of the first triangle. Cutting the two soft lines leaves a hard gauge-boson exchange on the left and a quartic vertex on the right, joined by the central soft loop factor. The second triangle follows by exchanging the two external pairs.

in (25). The two cubic vertices and the hard line instead define

$$\begin{aligned} \sum_{a,b=\pm 1} (-ab) \int_{-\infty}^0 \frac{d\tau_1 d\tau_2}{|H\tau_1|^2 |H\tau_2|^2} G_a(k_1; \tau_1) G_b(k_2; \tau_2) D_{ab}^{(\lambda)}(k_1; \tau_1, \tau_2) (\tau_1 \tau_2)^{1/2+i\eta\tilde{\nu}} \\ \simeq \frac{k_{12}^{-2i\eta\tilde{\nu}}}{4k_1^3 k_2^3} \mathcal{P}_\eta^{(2),\lambda}(\tilde{\nu}). \end{aligned} \quad (31)$$

At leading order, $k_1 \simeq k_2 \simeq k_{12}/2$. The integral therefore reduces to the double-folded scalar seeds of Ref. [3], which we denote by $\mathcal{M}_{ab}^{p_1 p_2} \equiv \tilde{\mathcal{I}}_{ab}^{p_1 p_2}(1, 1)$. For the transverse modes this follows directly from (7). For the longitudinal mode, the differential relation also produces a contact contribution [2]. We retain this term together with the scalar-seed contributions. The required seed exponents include $p_0 = -5/2 + i\eta\tilde{\nu}$, on the boundary of the convergence domain. We define the result by analytic continuation from $p_0 + \epsilon$, $\epsilon > 0$, and take the finite part of the assembled expression at $\epsilon = 0$. The seed normalization, closed forms, and combinations giving $\mathcal{P}_\eta^{(2),\lambda}$ are collected in App. C.

Finally, the leading signal of the first triangle is

$$\left[\langle \delta h^4 \rangle'_{\text{Tri1}} \right]_{\text{NonL}} \simeq \frac{g^6 h_0^2 H^2 k_s^3}{128 k_1^3 k_2^3 k_3^3 k_4^3} \left[\left(\frac{k_s^2}{4k_{12}k_{34}} \right)^{2i\tilde{\nu}} C^{\text{Tri}}(\tilde{\nu}) + \text{c.c.} \right], \quad (32)$$

where

$$C^{\text{Tri}}(\tilde{\nu}) = \frac{\Gamma^4(-i\tilde{\nu})}{(4\pi)^{7/2}} \mathcal{P}_+^{(1)}(\tilde{\nu}) \sum_{\lambda=L,\pm} \mathcal{P}_+^{(2),\lambda}(\tilde{\nu}) \mathcal{I}_{\text{Tri}}^\lambda(-i\tilde{\nu}, \theta). \quad (33)$$

Since $\mathcal{P}_\eta^{(2),+} = \mathcal{P}_\eta^{(2),-}$, the two transverse helicities contribute equally. The simplified loop coefficients make C^{Tri} independent of θ at this order; its reduced expression is given in (52).

The second triangle has the quartic vertex on the (1, 2) side and follows by $(1, 2) \leftrightarrow (3, 4)$. Adding the two placements gives

$$\left[\langle \delta h^4 \rangle'_{\text{Tri}} \right]_{\text{NonL}} \simeq \frac{g^6 h_0^2 H^2 k_s^3}{128 k_1^3 k_2^3 k_3^3 k_4^3} \left\{ \left(\frac{k_s^2}{4k_{12}k_{34}} \right)^{2i\tilde{\nu}} \left[C^{\text{Tri}}(\tilde{\nu}) + C^{\text{Tri}}(\tilde{\nu}) \right] + \text{c.c.} \right\}. \quad (34)$$

The nonanalytic momentum dependence is the same as in the bubble, while the hard propagator changes the amplitude and phase. Figure 5 shows the coefficient of a single triangle.

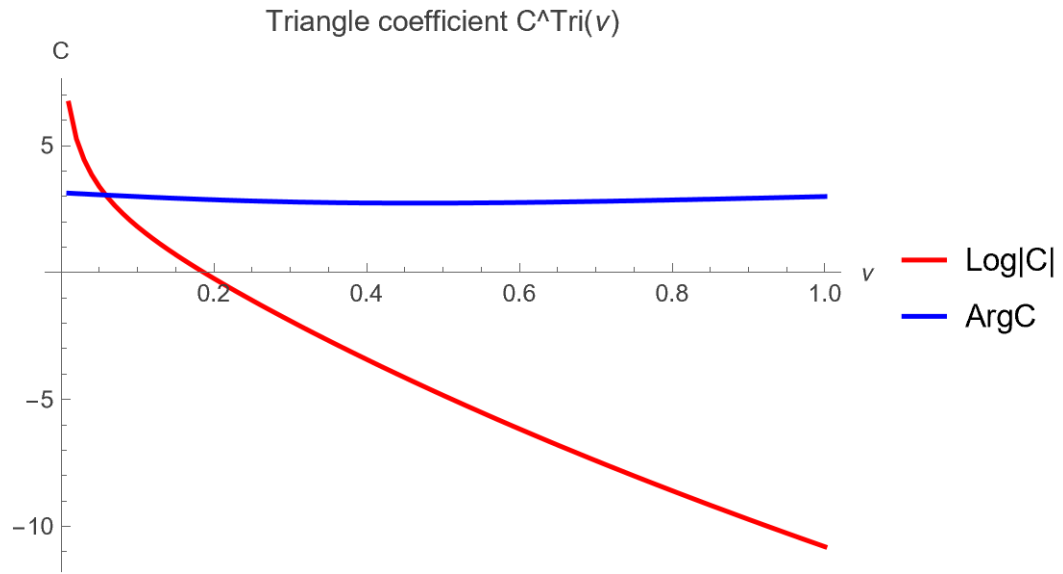


Figure 5: The coefficient $C^{\text{Tri}}(\tilde{\nu})$ defined in (33). The red and blue curves show $\log |C^{\text{Tri}}|$ and $\arg C^{\text{Tri}}$ in radians, respectively. The horizontal variable is $\tilde{\nu}$.

2.4 Box Diagram

Now we consider the Higgs trispectrum from the box diagram. The four cubic vertices give the coupling factor $(g^2 h_0/2)^4 = g^8 h_0^4/16$, and the diagram symmetry factor is unity.

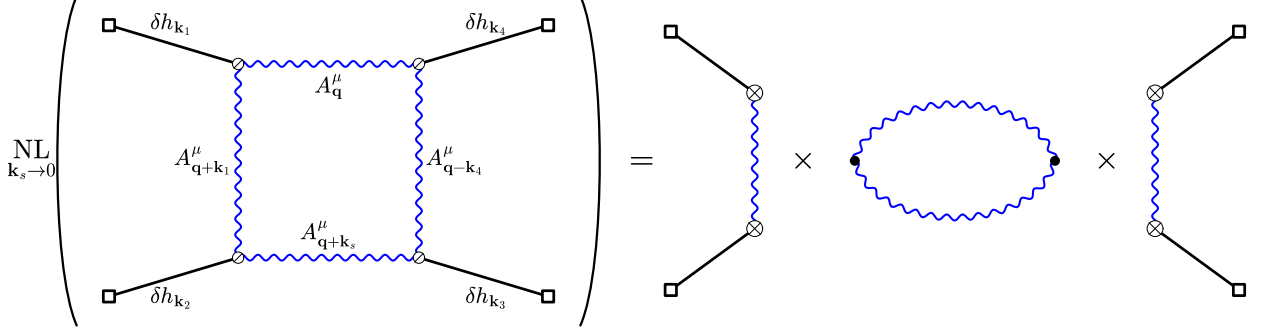


Figure 6: The leading nonlocal part of the box. Cutting the two soft lines separates two tree subgraphs, each containing one hard gauge-boson propagator. The central factor contains the nonlocal parts of the two soft propagators and their loop momentum integral.

Choose the loop momenta along the four successive internal lines as $\mathbf{q} + \mathbf{k}_1$, $\mathbf{K} = \mathbf{q} + \mathbf{k}_s$, $\mathbf{q} - \mathbf{k}_4$, and $\mathbf{q}$. In the collapsed limit, the leading nonlocal signal comes from the region $q, K \sim k_s$. Cutting these two soft lines leaves two hard propagators, with momenta approaching $\mathbf{k}_1$ and $\mathbf{k}_3$. Their polarizations are denoted by $\lambda_1, \lambda_3 = L, \pm$. The soft loop gives the dimensionless coefficient $\mathcal{I}_{Box}^{\lambda_1 \lambda_3}(-i\eta\tilde{\nu}, \theta_1, \theta_3, \phi_3)$ defined in (58). Its common momentum factor is $4^{-2i\eta\tilde{\nu}} k_s^{3+4i\eta\tilde{\nu}}$, as for the bubble and triangle. As shown in Fig. 6, the time integrals factorize into two copies of (31), one for each hard line. They contribute $\mathcal{P}_\eta^{(2), \lambda_1}(\tilde{\nu})$ and $\mathcal{P}_\eta^{(2), \lambda_3}(\tilde{\nu})$. Combining the two branches gives

$$\left[\langle \delta h^4 \rangle'_{Box} \right]_{NonL} \simeq \frac{g^8 h_0^4 k_s^3}{256 k_1^3 k_2^3 k_3^3 k_4^3} \left[\left(\frac{k_s^2}{4 k_{12} k_{34}} \right)^{2i\tilde{\nu}} \times C^{Box}(\tilde{\nu}, \theta_1, \theta_3, \phi_3) + \text{c.c.} \right], \quad (35)$$

where

$$C^{Box}(\tilde{\nu}, \theta_1, \theta_3, \phi_3) = \frac{\Gamma^4(-i\tilde{\nu})}{(4\pi)^{7/2}} \sum_{\lambda_1, \lambda_3 = L, \pm} \mathcal{P}_+^{(2), \lambda_1}(\tilde{\nu}) \mathcal{P}_+^{(2), \lambda_3}(\tilde{\nu}) \times \mathcal{I}_{Box}^{\lambda_1 \lambda_3}(-i\tilde{\nu}, \theta_1, \theta_3, \phi_3). \quad (36)$$

Since the hard transverse propagators have equal time integrals, the polarization sum reduces to the LL , LH , and HH groups in (59). Their explicit expressions are given in App. D. In particular, (65) shows that all angular dependence is carried by $\mathcal{I}_{Box}^{LL}$ and multiplied by $[\mathcal{P}_+^{(2), L} - \mathcal{P}_+^{(2), +}]^2$. The difference between the longitudinal and transverse hard-line time integrals therefore controls the angular dependence of the box signal.

The result contains five angular structures:

$$\begin{aligned} C^{Box}(\tilde{\nu}, \theta_1, \theta_3, \phi_3) = & A_1(\tilde{\nu}) + A_2(\tilde{\nu})(\cos^2 \theta_1 + \cos^2 \theta_3) + A_3(\tilde{\nu}) \cos^2 \theta_1 \cos^2 \theta_3 \\ & + B(\tilde{\nu}) \sin(2\theta_1) \sin(2\theta_3) \cos \phi_3 \\ & + D(\tilde{\nu}) \sin^2 \theta_1 \sin^2 \theta_3 \cos(2\phi_3). \end{aligned} \quad (37)$$

The five complex functions depend only on $\tilde{\nu}$. The signal is symmetric under $\theta_1 \leftrightarrow \theta_3$ and $\phi_3 \rightarrow -\phi_3$. Its oscillatory momentum dependence is the same as in (28), with an angle-dependent amplitude and phase supplied by C^{Box} .

Figures 7 and 8 show the five coefficient functions. Their magnitudes decrease over the displayed mass range. The jumps in the plotted phases reflect the principal branch of $\arg$, defined modulo 2π .

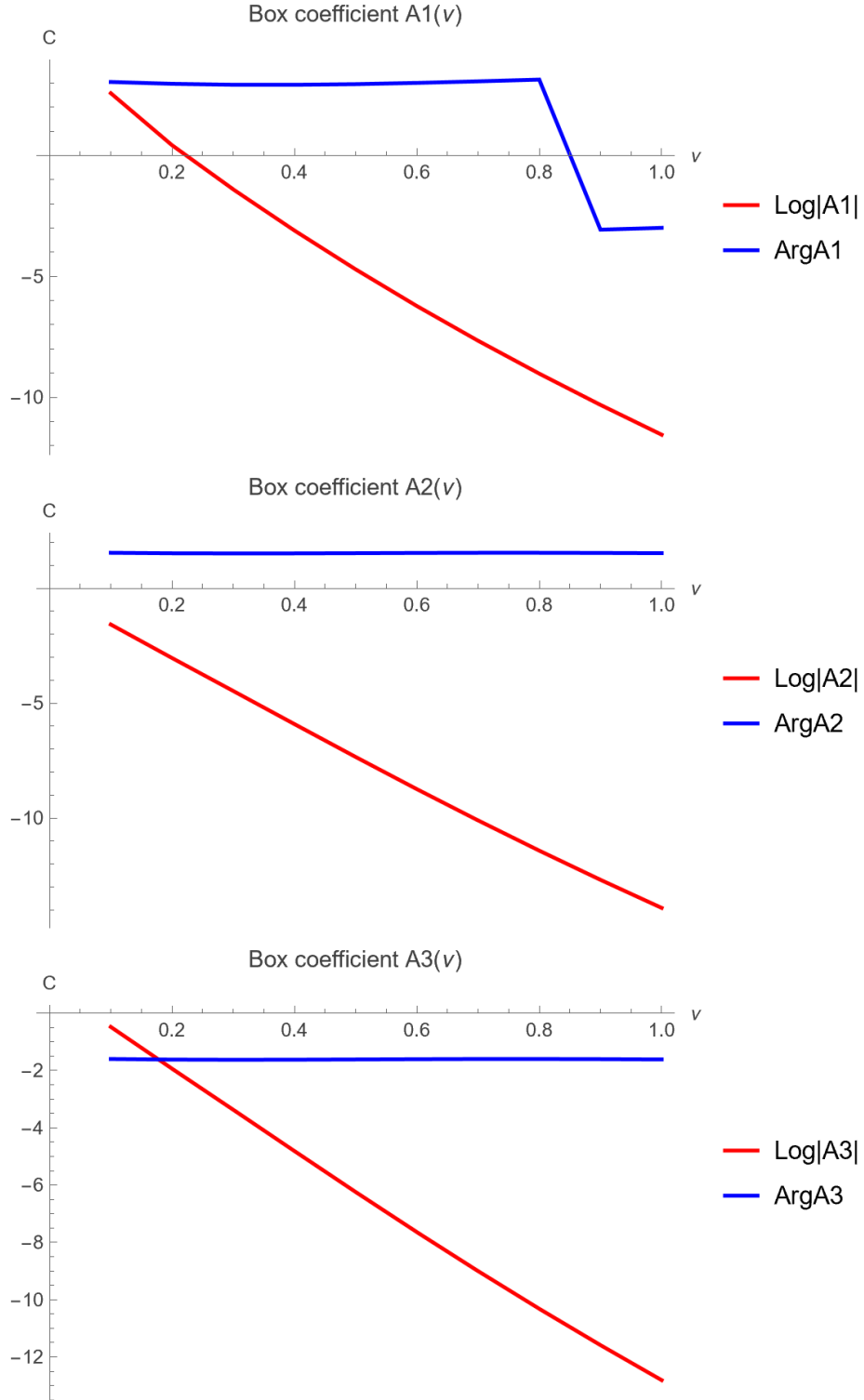


Figure 7: The angular coefficients A_1 , A_2 , and A_3 (top to bottom) in (37) for the box diagram. Red curves show the logarithmic magnitudes and blue curves show the principal phases in radians. The horizontal variable is $\tilde{\nu}$.

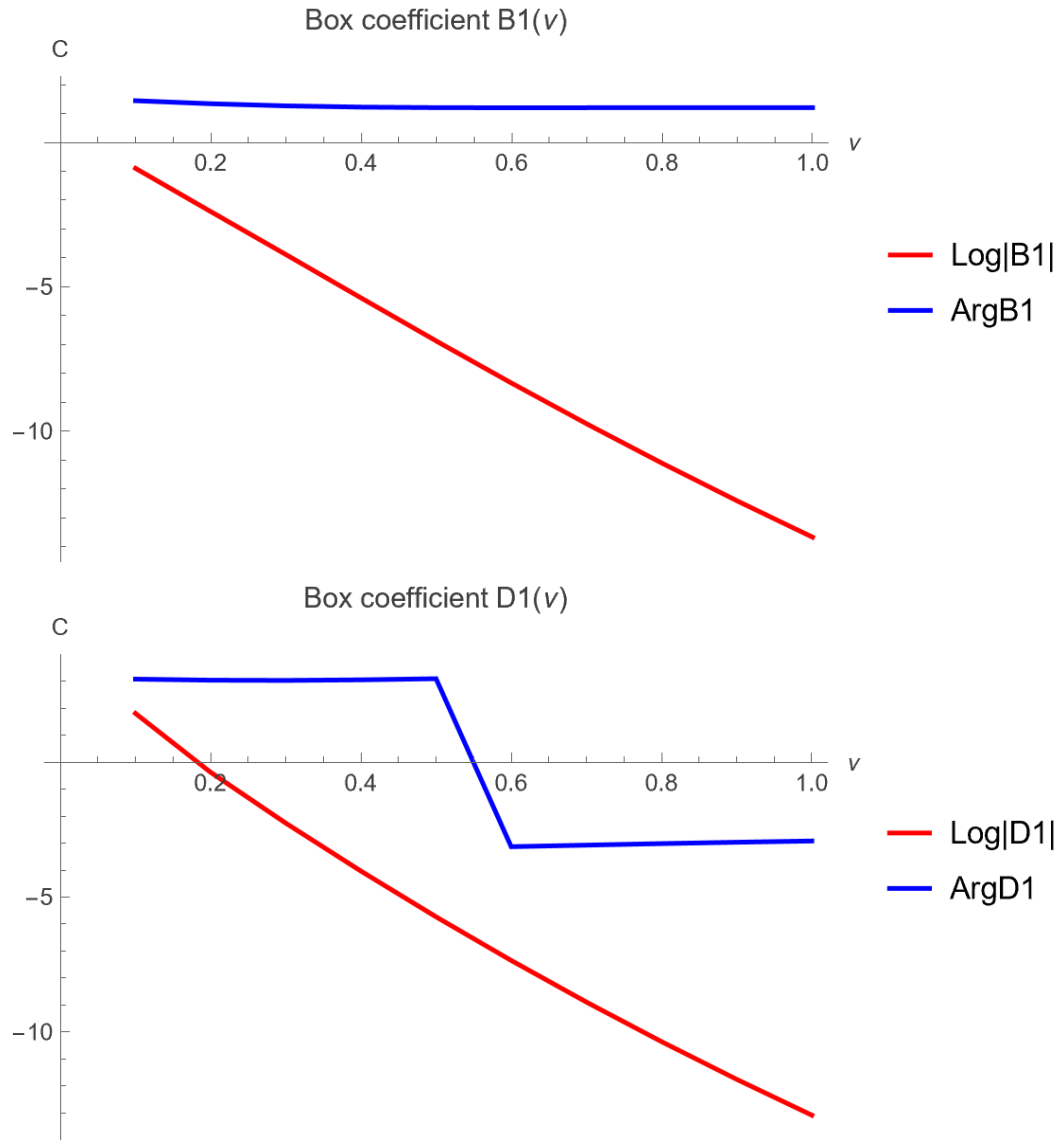


Figure 8: The angular coefficients B (top, labelled $B1$) and D (bottom, labelled $D1$) in (37). The plotting conventions are the same as in Fig. 7.

3 Top Quark Signal

4 Conclusions

A Loop Integral Formulae

The identity $z = (K^2 - q^2 - k_s^2)/(2qk_s)$ reduces all angular moments in (15) to the scalar integral f_0 :

$$f_r(\alpha, \beta) = \frac{1}{2^r} \sum_{u+v+w=r} \frac{r!}{u!v!w!} (-1)^{v+w} f_0\left(\alpha + \frac{r}{2} - v, \beta - u\right), \quad (38)$$

where the sum runs over nonnegative integers. For the third and fourth moments this gives

$$\begin{aligned} f_3(\alpha, \beta) = \frac{1}{8} & \left[f_0\left(\alpha + \frac{3}{2}, \beta - 3\right) - 3f_0\left(\alpha + \frac{3}{2}, \beta - 2\right) - 3f_0\left(\alpha + \frac{1}{2}, \beta - 2\right) \right. \\ & + 3f_0\left(\alpha + \frac{3}{2}, \beta - 1\right) + 6f_0\left(\alpha + \frac{1}{2}, \beta - 1\right) + 3f_0\left(\alpha - \frac{1}{2}, \beta - 1\right) \\ & \left. - f_0\left(\alpha + \frac{3}{2}, \beta\right) - 3f_0\left(\alpha + \frac{1}{2}, \beta\right) - 3f_0\left(\alpha - \frac{1}{2}, \beta\right) - f_0\left(\alpha - \frac{3}{2}, \beta\right) \right], \end{aligned} \quad (39)$$

$$\begin{aligned} f_4(\alpha, \beta) = \frac{1}{16} & \left[f_0(\alpha + 2, \beta - 4) - 4f_0(\alpha + 2, \beta - 3) - 4f_0(\alpha + 1, \beta - 3) \right. \\ & + 6f_0(\alpha + 2, \beta - 2) + 12f_0(\alpha + 1, \beta - 2) + 6f_0(\alpha, \beta - 2) \\ & - 4f_0(\alpha + 2, \beta - 1) - 12f_0(\alpha + 1, \beta - 1) - 12f_0(\alpha, \beta - 1) \\ & - 4f_0(\alpha - 1, \beta - 1) + f_0(\alpha + 2, \beta) + 4f_0(\alpha + 1, \beta) \\ & \left. + 6f_0(\alpha, \beta) + 4f_0(\alpha - 1, \beta) + f_0(\alpha - 2, \beta) \right]. \end{aligned} \quad (40)$$

B Bubble Coefficients

For $x = -i\eta\tilde{\nu}$, the relative weight of a longitudinal nonlocal propagator is

$$\beta(x) \equiv \frac{(x + 1/2)^2}{1/4 - x^2} = \frac{1 + 2x}{1 - 2x}. \quad (41)$$

Define $\mathcal{I}_{\sigma_1\sigma_2}(x)$ by restricting the loop in (22) to the polarizations (σ_1, σ_2) and replacing $\mathcal{I}_{Bub}$ on its right-hand side by $\mathcal{I}_{\sigma_1\sigma_2}$. The independent coefficients are

$$\mathcal{I}_{LL}(x) = [\beta(x)]^2 [\mathcal{J}_0(2, 2; x) + 2\mathcal{J}_1(1, 2; x) + \mathcal{J}_2(0, 2; x)], \quad (42)$$

$$\mathcal{I}_{L+}(x) = \frac{\beta(x)}{2} [\mathcal{J}_0(0, 2; x) - \mathcal{J}_2(0, 2; x)], \quad (43)$$

$$\begin{aligned} \mathcal{I}_{++}(x) = \frac{1}{4} & [\mathcal{J}_0(0, 0; x) + \mathcal{J}_0(2, 2; x) + \mathcal{J}_2(0, 2; x) \\ & + 2\mathcal{J}_0(1, 1; x) + 2\mathcal{J}_1(1, 2; x) + 2\mathcal{J}_1(0, 1; x)], \end{aligned} \quad (44)$$

$$\begin{aligned} \mathcal{I}_{+-}(x) = \frac{1}{4} & [\mathcal{J}_0(0, 0; x) + \mathcal{J}_0(2, 2; x) + \mathcal{J}_2(0, 2; x) \\ & - 2\mathcal{J}_0(1, 1; x) + 2\mathcal{J}_1(1, 2; x) - 2\mathcal{J}_1(0, 1; x)]. \end{aligned} \quad (45)$$

The remaining entries satisfy $\mathcal{I}_{--} = \mathcal{I}_{++}$, $\mathcal{I}_{-+} = \mathcal{I}_{+-}$, and $\mathcal{I}_{L-} = \mathcal{I}_{+L} = \mathcal{I}_{-L} = \mathcal{I}_{L+}$. Consequently,

$$\mathcal{I}_{Bub} = \mathcal{I}_{LL} + 4\mathcal{I}_{L+} + 2\mathcal{I}_{++} + 2\mathcal{I}_{+-}, \quad (46)$$

which reduces to (23) using the moment relations in App. A.

For the $\eta = +1$ branch, the fully reduced sum is

$$\mathcal{I}_{Bub}(-i\tilde{\nu}) = \frac{3 \cdot 2^{-4-4i\tilde{\nu}} (3i - 2\tilde{\nu})^2 \Gamma(-3/2 - 2i\tilde{\nu}) \Gamma(3 + 2i\tilde{\nu}) \sinh^2(\pi\tilde{\nu})}{\pi[1 + (3i - 2\tilde{\nu})\tilde{\nu}]^2}. \quad (47)$$

Substituting this into (27) gives

$$\begin{aligned} C^{Bub}(\tilde{\nu}) &= \frac{3(\tilde{\nu} - i)(\tilde{\nu} - 2i)^4}{256\pi^{7/2}\tilde{\nu}^2(2\tilde{\nu} - i)} \Gamma^2(-2 - i\tilde{\nu}) \Gamma^2(5/2 + i\tilde{\nu}) \\ &\quad \times \Gamma(-3/2 - 2i\tilde{\nu}) \Gamma(1 + 2i\tilde{\nu}) \sinh^2(\pi\tilde{\nu}). \end{aligned} \quad (48)$$

For real $\tilde{\nu} > 0$, the opposite nonlocal branch is their complex conjugate.

C Triangle Coefficients

C.1 Polarization sum

Let $x = -i\eta\tilde{\nu}$ and let $\beta(x)$ be the longitudinal weight in (41). Summing the nine soft-polarization combinations gives

$$\begin{aligned} \mathcal{I}_{Tri}^L(x, \theta) &= [\beta(x)]^2 \left\{ \frac{\sin^2 \theta}{2} [\mathcal{J}_0(2, 2; x) + \mathcal{J}_1(1, 2; x) - \mathcal{J}_2(2, 2; x) - \mathcal{J}_3(1, 2; x)] \right. \\ &\quad \left. + \cos^2 \theta [\mathcal{J}_2(2, 2; x) + \mathcal{J}_3(1, 2; x) + \mathcal{J}_1(1, 2; x) + \mathcal{J}_2(0, 2; x)] \right\} \\ &\quad + \frac{\beta(x)(1 + \cos^2 \theta)}{2} [\mathcal{J}_0(0, 2; x) - \mathcal{J}_2(0, 2; x)] \\ &\quad + \frac{\sin^2 \theta}{2} [\mathcal{J}_2(2, 2; x) + \mathcal{J}_3(1, 2; x) + \mathcal{J}_1(1, 2; x) + \mathcal{J}_2(0, 2; x) + \mathcal{J}_0(0, 0; x)] \\ &\quad + \cos^2 \theta [\mathcal{J}_0(2, 2; x) - \mathcal{J}_2(2, 2; x) + \mathcal{J}_1(1, 2; x) - \mathcal{J}_3(1, 2; x)]. \end{aligned} \quad (49)$$

The terms proportional to β^2 , β , and 1 arise from two longitudinal, mixed, and two transverse soft lines, respectively.

For a hard line with helicity +, the corresponding expression is

$$\begin{aligned} \mathcal{I}_{Tri}^+(x, \theta) &= [\beta(x)]^2 \left\{ \frac{1 + \cos^2 \theta}{4} [\mathcal{J}_0(2, 2; x) - \mathcal{J}_2(2, 2; x) + \mathcal{J}_1(1, 2; x) - \mathcal{J}_3(1, 2; x)] \right. \\ &\quad \left. + \frac{\sin^2 \theta}{2} [\mathcal{J}_2(2, 2; x) + \mathcal{J}_1(1, 2; x) + \mathcal{J}_3(1, 2; x) + \mathcal{J}_2(0, 2; x)] \right\} \\ &\quad + \frac{\beta(x)(2 + \sin^2 \theta)}{4} [\mathcal{J}_0(0, 2; x) - \mathcal{J}_2(0, 2; x)] \\ &\quad + \frac{1 + \cos^2 \theta}{4} [\mathcal{J}_2(2, 2; x) + \mathcal{J}_1(1, 2; x) + \mathcal{J}_3(1, 2; x) + \mathcal{J}_2(0, 2; x) + \mathcal{J}_0(0, 0; x)] \\ &\quad + \frac{\sin^2 \theta}{2} [\mathcal{J}_0(2, 2; x) - \mathcal{J}_2(2, 2; x) + \mathcal{J}_1(1, 2; x) - \mathcal{J}_3(1, 2; x)]. \end{aligned} \quad (50)$$

At zero chemical potential, reducing the Gamma functions gives the same result for all three hard-line polarizations:

$$\begin{aligned}\mathcal{I}_{Tri}^L(-i\tilde{\nu}, \theta) &= \mathcal{I}_{Tri}^+(-i\tilde{\nu}, \theta) = \mathcal{I}_{Tri}^-(-i\tilde{\nu}, \theta) = \frac{1}{3}\mathcal{I}_{Bub}(-i\tilde{\nu}) \\ &= \frac{2^{-4-4i\tilde{\nu}}(3i-2\tilde{\nu})^2\Gamma(-3/2-2i\tilde{\nu})\Gamma(3+2i\tilde{\nu})\sinh^2(\pi\tilde{\nu})}{\pi[1+(3i-2\tilde{\nu})\tilde{\nu}]^2}.\end{aligned}\quad (51)$$

Summing over the hard-line polarizations reproduces $\mathcal{I}_{Bub}$ by polarization completeness. Thus the angular dependence of the separate soft-polarization contributions cancels in their sum. The coefficient of a single triangle reduces to

$$C^{Tri}(\tilde{\nu}) = \frac{\Gamma^4(-i\tilde{\nu})}{3(4\pi)^{7/2}}\mathcal{P}_+^{(1)}(\tilde{\nu}) \left[\mathcal{P}_+^{(2),L}(\tilde{\nu}) + 2\mathcal{P}_+^{(2),+}(\tilde{\nu}) \right] \mathcal{I}_{Bub}(-i\tilde{\nu}). \quad (52)$$

C.2 Time-integral coefficients

For the scalar propagator of (6), the double-folded seed is defined by

$$\mathcal{M}_{ab}^{p_1 p_2} \equiv (-ab)k^{5+p_{12}} \int_{-\infty}^0 d\tau_1 d\tau_2 (-\tau_1)^{p_1} (-\tau_2)^{p_2} e^{iak\tau_1 + ibk\tau_2} \frac{D_{ab}^{(s)}(k; \tau_1, \tau_2)}{H^2}, \quad (53)$$

This is the seed $\tilde{\mathcal{I}}_{ab}^{p_1 p_2}(1, 1)$ of Ref. [3], with H restored explicitly. All seeds below have mass index $\tilde{\nu}$. The closed expressions are

$$\begin{aligned}\mathcal{M}_{\pm\mp}^{p_1 p_2} &= \frac{e^{\mp i\pi p_{12}/2}}{2^{5+p_{12}}} \prod_{j=1}^2 \frac{\Gamma(5/2+p_j-i\tilde{\nu})\Gamma(5/2+p_j+i\tilde{\nu})}{\Gamma(3+p_j)}, \\ \mathcal{M}_{\pm\pm}^{p_1 p_2} &= \frac{\pm ie^{\mp i\pi p_{12}/2} e^{-\pi\tilde{\nu}}}{2^{5+p_{12}}} \prod_{j=1}^2 \frac{\Gamma(5/2+p_j-i\tilde{\nu})\Gamma(5/2+p_j+i\tilde{\nu})}{\Gamma(3+p_j)} \\ &\quad - \frac{e^{\mp i\pi p_{12}/2}}{2^{5+p_{12}}} \Gamma(5+p_{12})\Gamma(5/2+p_1 \pm i\tilde{\nu})\Gamma(5/2+p_2 \pm i\tilde{\nu}) \\ &\quad \times {}_3\tilde{F}_2 \left[\begin{matrix} 5+p_{12}, 1/2 \pm i\tilde{\nu}, 1 \\ 7/2+p_1 \pm i\tilde{\nu}, 7/2+p_2 \pm i\tilde{\nu} \end{matrix} \middle| 1 \right].\end{aligned}\quad (54)$$

Here ${}_3\tilde{F}_2(a, b, c; d, e; 1) = {}_3F_2(a, b, c; d, e; 1)/[\Gamma(d)\Gamma(e)]$ is the regularized hypergeometric function.

Set $p_\epsilon = -5/2 + \epsilon + i\eta\tilde{\nu}$. The two transverse coefficients are

$$\mathcal{P}_\eta^{(2),\pm}(\tilde{\nu}) = 2^{2i\eta\tilde{\nu}} \text{FP}_{\epsilon=0} \sum_{a,b=\pm 1} \left[\mathcal{M}_{ab}^{p_\epsilon, p_\epsilon} + ia\mathcal{M}_{ab}^{p_\epsilon+1, p_\epsilon} + ib\mathcal{M}_{ab}^{p_\epsilon, p_\epsilon+1} - ab\mathcal{M}_{ab}^{p_\epsilon+1, p_\epsilon+1} \right]. \quad (56)$$

The longitudinal coefficient can be written using the same transverse combination:

$$\begin{aligned}\mathcal{P}_\eta^{(2),L}(\tilde{\nu}) &= \frac{\mathcal{P}_\eta^{(1)}(\tilde{\nu}) + (1/2 + i\eta\tilde{\nu})^2 \mathcal{P}_\eta^{(2),+}(\tilde{\nu})}{\tilde{\nu}^2 + 1/4} \\ &\quad + \frac{2^{2i\eta\tilde{\nu}}}{\tilde{\nu}^2 + 1/4} \text{FP}_{\epsilon=0} \sum_{a,b=\pm 1} \left\{ (1/2 + i\eta\tilde{\nu}) \left[\mathcal{M}_{ab}^{p_\epsilon+2, p_\epsilon} + ib\mathcal{M}_{ab}^{p_\epsilon+2, p_\epsilon+1} \right. \right. \\ &\quad \left. \left. + \mathcal{M}_{ab}^{p_\epsilon, p_\epsilon+2} + ia\mathcal{M}_{ab}^{p_\epsilon+1, p_\epsilon+2} \right] + \mathcal{M}_{ab}^{p_\epsilon+2, p_\epsilon+2} \right\}.\end{aligned}\quad (57)$$

The operator $\text{FP}_{\epsilon=0}$ selects the constant coefficient of the Laurent expansion after assembling the indicated seed sum. In particular, the seed expressions are combined before setting ϵ to zero. The first term proportional to $\mathcal{P}_\eta^{(1)}$ in (57) is the contact contribution from differentiating the time-ordering step functions.

D Box Coefficients

D.1 Polarization sums

Let $\lambda_1, \lambda_3 = L, \pm$ denote the polarizations of the hard lines with momenta $\mathbf{k}_1$ and $\mathbf{k}_3$. With $\mathbf{K} = \mathbf{q} + \mathbf{k}_s$, define the soft-polarization sum by

$$\begin{aligned} \sum_{\sigma_1, \sigma_2=L, \pm} [\beta(x)]^{\delta_{\sigma_1 L} + \delta_{\sigma_2 L}} \int \frac{d^3 \mathbf{q}}{(2\pi)^3} \left(\frac{q}{2}\right)^{-2x} \left(\frac{K}{2}\right)^{-2x} (e_{\mathbf{k}_1}^{(\lambda_1)*} \cdot e_{\mathbf{K}}^{(\sigma_1)}) (e_{\mathbf{K}}^{(\sigma_1)*} \cdot e_{\mathbf{k}_3}^{(\lambda_3)}) \\ \times (e_{\mathbf{k}_3}^{(\lambda_3)*} \cdot e_{\mathbf{q}}^{(\sigma_2)}) (e_{\mathbf{q}}^{(\sigma_2)*} \cdot e_{\mathbf{k}_1}^{(\lambda_1)}) = \frac{4^{2x} k_s^{3-4x}}{(4\pi)^{3/2}} \mathcal{I}_{\text{Box}}^{\lambda_1 \lambda_3}(x, \theta_1, \theta_3, \phi_3). \end{aligned} \quad (58)$$

The angles are defined by the convention stated below (1). The two transverse hard-line coefficients $\mathcal{P}_\eta^{(2),+} = \mathcal{P}_\eta^{(2),-}$ are equal, so it suffices to retain three groups:

$$\mathcal{I}_{\text{Box}}^{LH} \equiv \sum_{\lambda=\pm} \left(\mathcal{I}_{\text{Box}}^{L\lambda} + \mathcal{I}_{\text{Box}}^{\lambda L} \right), \quad \mathcal{I}_{\text{Box}}^{HH} \equiv \sum_{\lambda_1, \lambda_3=\pm} \mathcal{I}_{\text{Box}}^{\lambda_1 \lambda_3}, \quad (59)$$

together with $\mathcal{I}_{\text{Box}}^{LL}$. Here H in a polarization superscript denotes the sum over transverse helicities.

For $x = -i\tilde{\nu}$, the longitudinal result is

$$\begin{aligned} \mathcal{I}_{\text{Box}}^{LL}(-i\tilde{\nu}, \theta_1, \theta_3, \phi_3) \\ = - \frac{4^{-3-2i\tilde{\nu}} (2\tilde{\nu} - 3i) \Gamma(-3/2 - 2i\tilde{\nu}) \Gamma(1 + 2i\tilde{\nu}) \sinh^2(\pi\tilde{\nu})}{\pi(\tilde{\nu} - i)(\tilde{\nu} - 2i)(2\tilde{\nu} - i)} \\ \times \left\{ -16 - 19i\tilde{\nu} + 6\tilde{\nu}^2 + \tilde{\nu}(2\tilde{\nu} - i) [\cos(2\theta_1) + \cos(2\theta_3) + 3\cos(2\theta_1)\cos(2\theta_3)] \right. \\ \left. + 4(2\tilde{\nu}^2 - 5i\tilde{\nu} - 3) \cos(2\phi_3) \sin^2 \theta_1 \sin^2 \theta_3 \right. \\ \left. - 8\tilde{\nu}(\tilde{\nu} - i) \cos \phi_3 \sin(2\theta_1) \sin(2\theta_3) \right\}. \end{aligned} \quad (60)$$

Reducing the mixed and transverse results to the same Gamma-function basis gives

$$\begin{aligned}
& \mathcal{I}_{\text{Box}}^{LH}(-i\tilde{\nu}, \theta_1, \theta_3, \phi_3) \\
&= -\frac{2 \cdot 4^{-3-2i\tilde{\nu}}(2\tilde{\nu} - 3i)\Gamma(-3/2 - 2i\tilde{\nu})\Gamma(1 + 2i\tilde{\nu})\sinh^2(\pi\tilde{\nu})}{\pi(\tilde{\nu} - i)(\tilde{\nu} - 2i)(2\tilde{\nu} - i)} \\
&\times \left\{ -32 - 37i\tilde{\nu} + 10\tilde{\nu}^2 - \tilde{\nu}(2\tilde{\nu} - i) \left[\cos(2\theta_1) + \cos(2\theta_3) + 3\cos(2\theta_1)\cos(2\theta_3) \right] \right. \\
&\quad - 4(2\tilde{\nu}^2 - 5i\tilde{\nu} - 3)\cos(2\phi_3)\sin^2\theta_1\sin^2\theta_3 \\
&\quad \left. + 8\tilde{\nu}(\tilde{\nu} - i)\cos\phi_3\sin(2\theta_1)\sin(2\theta_3) \right\}, \tag{61}
\end{aligned}$$

$$\begin{aligned}
& \mathcal{I}_{\text{Box}}^{HH}(-i\tilde{\nu}, \theta_1, \theta_3, \phi_3) \\
&= -\frac{4^{-3-2i\tilde{\nu}}(2\tilde{\nu} - 3i)\Gamma(-3/2 - 2i\tilde{\nu})\Gamma(1 + 2i\tilde{\nu})\sinh^2(\pi\tilde{\nu})}{\pi(\tilde{\nu} - i)(\tilde{\nu} - 2i)(2\tilde{\nu} - i)} \\
&\times \left\{ -64 - 75i\tilde{\nu} + 22\tilde{\nu}^2 + \tilde{\nu}(2\tilde{\nu} - i) \left[\cos(2\theta_1) + \cos(2\theta_3) + 3\cos(2\theta_1)\cos(2\theta_3) \right] \right. \\
&\quad + 4(2\tilde{\nu}^2 - 5i\tilde{\nu} - 3)\cos(2\phi_3)\sin^2\theta_1\sin^2\theta_3 \\
&\quad \left. - 8\tilde{\nu}(\tilde{\nu} - i)\cos\phi_3\sin(2\theta_1)\sin(2\theta_3) \right\}. \tag{62}
\end{aligned}$$

These expressions satisfy the relations implied by polarization completeness and (51):

$$\mathcal{I}_{\text{Box}}^{LH} = \frac{2}{3}\mathcal{I}_{\text{Bub}} - 2\mathcal{I}_{\text{Box}}^{LL}, \tag{63}$$

$$\mathcal{I}_{\text{Box}}^{HH} = \frac{1}{3}\mathcal{I}_{\text{Bub}} + \mathcal{I}_{\text{Box}}^{LL}. \tag{64}$$

Indeed, all angular terms cancel in $\mathcal{I}_{\text{Box}}^{LH} + 2\mathcal{I}_{\text{Box}}^{LL}$ and $\mathcal{I}_{\text{Box}}^{HH} - \mathcal{I}_{\text{Box}}^{LL}$. The three groups therefore also satisfy $\mathcal{I}_{\text{Box}}^{LL} + \mathcal{I}_{\text{Box}}^{LH} + \mathcal{I}_{\text{Box}}^{HH} = \mathcal{I}_{\text{Bub}}$.

The dimensionless box coefficient is consequently

$$\begin{aligned}
C^{\text{Box}}(\tilde{\nu}, \theta_1, \theta_3, \phi_3) &= \frac{\Gamma^4(-i\tilde{\nu})}{(4\pi)^{7/2}} \left\{ [\mathcal{P}_+^{(2),L}]^2 \mathcal{I}_{\text{Box}}^{LL} + \mathcal{P}_+^{(2),L} \mathcal{P}_+^{(2),+} \mathcal{I}_{\text{Box}}^{LH} + [\mathcal{P}_+^{(2),+}]^2 \mathcal{I}_{\text{Box}}^{HH} \right\} \\
&= \frac{\Gamma^4(-i\tilde{\nu})}{(4\pi)^{7/2}} \left\{ [\mathcal{P}_+^{(2),L} - \mathcal{P}_+^{(2),+}]^2 \mathcal{I}_{\text{Box}}^{LL} + \frac{1}{3} \mathcal{P}_+^{(2),+} [2\mathcal{P}_+^{(2),L} + \mathcal{P}_+^{(2),+}] \mathcal{I}_{\text{Bub}} \right\}, \tag{65}
\end{aligned}$$

The opposite nonlocal branch follows by complex conjugation.

D.2 Angular projections

The five functions in (37) can be read off from (65), or equivalently obtained from the following projections:

$$A_1(\tilde{\nu}) = \frac{1}{2} \left[C^{\text{Box}}(\tilde{\nu}, \frac{\pi}{2}, \frac{\pi}{2}, 0) + C^{\text{Box}}(\tilde{\nu}, \frac{\pi}{2}, \frac{\pi}{2}, \frac{\pi}{2}) \right], \quad (66)$$

$$A_2(\tilde{\nu}) = C^{\text{Box}}(\tilde{\nu}, \frac{\pi}{2}, 0, 0) - A_1(\tilde{\nu}), \quad (67)$$

$$A_3(\tilde{\nu}) = C^{\text{Box}}(\tilde{\nu}, 0, 0, 0) - A_1(\tilde{\nu}) - 2A_2(\tilde{\nu}), \quad (68)$$

$$B(\tilde{\nu}) = \frac{1}{2} \left[C^{\text{Box}}(\tilde{\nu}, \frac{\pi}{4}, \frac{\pi}{4}, 0) - C^{\text{Box}}(\tilde{\nu}, \frac{\pi}{4}, \frac{\pi}{4}, \pi) \right], \quad (69)$$

$$D(\tilde{\nu}) = \frac{1}{2} \left[C^{\text{Box}}(\tilde{\nu}, \frac{\pi}{2}, \frac{\pi}{2}, 0) - C^{\text{Box}}(\tilde{\nu}, \frac{\pi}{2}, \frac{\pi}{2}, \frac{\pi}{2}) \right]. \quad (70)$$

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